

INPUT

Lagrangian
configuration

Lagrangian
Builder

Exclusion
configuration

1 Viable-region search via "Battleship"

Repeat for B Rounds

Pheno tool suite

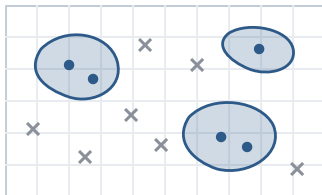
SModels
MICROMEGAS
⋮

"Shot" θ_t in
parameter
space

hit or miss?
exclusion
channels

LLaM

Large Lagrangian Model
autoregressive transformer
that reads a tokenized $\mathcal{L}_{\text{BSM}}$
and the search history



Viable regions $R_1 \dots R_n$

Disconnected sets in θ -space not
excluded by any experiment we
consider.

2 Signature decision tree

$R_1 \dots R_n$

Signal strength X
at expt. A?

yes

no

Signal strength Y
at expt. B?

Signal strength Z
at expt. C?

R_1

R_4

R_2

R_3

R_5

observables, computed by trusted
community-built tools

Competing LLM
agents

O^* ?

yes

no

R_3

R_5

OUTPUT

Desired Outcome

Unexplored observable O^*

e.g. halo-independent kinematic ratio applied to
paleo-detectors